

Urban Aerial Hierarchical Swarm Governance

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1. Project Objective & Scenario

Design and simulate a hierarchical traffic management system for autonomous drones operating in crowded urban airspace, where vehicles and a DTC make decisions using information exchanged through multiple communication paths.

Vehicles should use information from nearby vehicles to make immediate traffic and safety decisions, while the DTC manages larger-scale decisions such as traffic rules, flight corridors, priorities, and over-all traffic flow.

The system should define clear rules for when information is exchanged, which vehicle or system should make a decision, and how information is passed between vehicles and the DTC.

This may include direct vehicle-to-vehicle communication, vehicle-to-DTC communication, DTC-to-vehicle communication, and decisions that combine information from multiple vehicles before reaching the DTC.

The system must support different vehicle priorities and continue to operate safely when communication is limited or when the DTC is unavailable. It should also remain robust when some vehicles provide false information.

The system should maintain safe separation between vehicles, resolve conflicts, ensure that vehicles reach their destinations, and prevent individual faulty or malicious vehicles from disrupting the over-all traffic.

2. Scope & Expected Deliverables

- **Simulation Environment:** The system will be demonstrated in a 2.5D simulated urban airspace with predefined corridors, intersections, entry/exit points, and obstacles. The simulation will use a finite number of heterogeneous drone agents with different priorities, speeds, destinations, and communication capabilities. Vehicles will be represented primarily by their position, velocity, destination, and priority rather than detailed physical models.
- **Traffic Conditions:** Experiments will consider cases of low, medium, and high traffic densities, including corridor merging and conflicting traffic flows. The environment will remain bounded and use predefined routes/corridors rather than modeling an entire city.
- **Information Flow:** The study will consider drone-to-drone, drone-to-overseer, and overseer-to-drone communication paths, including multi-step information exchange. Communication will be modeled through real-time communication.
- **Fault and Adversarial Conditions:** The experiments will focus on position falsification, incorrect intent, rule violations, and communication loss as representative faulty/adversarial behaviours. More complex cyberattacks and real-world hardware failures are outside the scope.
- **Global Overseer:** The overseer will provide high-level traffic decisions such as corridor allocation and priority decisions rather than continuously controlling individual drone movements.
- **Safety:** Safety evaluation will focus primarily on minimum separation, collision avoidance, and successful destination arrival. Physical aircraft safety and real-world regulatory certification are outside the scope.

(Note: Advanced features like true 3D rendering, 500+ agents, full epistemic-logic belief modeling are excluded from the core commitments to ensure feasible execution, but may be explored as extra goals if time permits).

3. Individual Contributions

The technical workload is divided as follows:

- **Abdur Rehman & Dhadheechi** : Building the live simulation world, including agent dynamics, the communication graph, rendering, and integration of all modules, while also formalizing safety requirements as Signal Temporal Logic (STL) specifications, implementing safety-spec monitoring, and assembling the final dashboard and visualization.
- **Lalith Vishnu**: Designing and implementing the distributed optimization layer using Alternating Direction Method of Multipliers (ADMM) to resolve conflicts involving yielding, merging, and rerouting between agents.
- **Bhuvan** : Implementing Boids-style local rules and corridor/platoon merging logic, covering local alignment, cohesion, separation, and autonomous formation behavior.
- **Tanmay** : Developing the centralized overseer node responsible for corridor allocation and priority arbitration across different drone types, using global traffic rules and priority ordering.
- **Sampadram** : Implementing fault and attack injection together with W-MSR-style resilient consensus to defend against malicious or faulty agents and maintain resilient swarm behavior.

4. Demonstration Vision

The project will culminate in a real-time visual demonstration consisting of a dual screen dashboard to clearly prove the architecture's capabilities:

- **Panel 1 (Live Simulation)**: A top-down view of the airspace featuring drones represented as colored dots using different colors for each type, navigating visible corridor lanes. The simulation will visually flag malicious drones when they broadcast bad data, and display the overseer's current directives as a subtle visual overlay.
- **Panel 2 (Live Dashboard)**: A real-time metrics dashboard featuring an STL robustness signal, a priority-compliance counter ticking up when emergency drones secure right-of-way, and a live timer tracking ADMM conflict-resolution convergence.
- **Visual Demonstration**: The presentation will walk through normal corridor formation, simultaneously inject a malicious drone while dropping the overseer link to show how the system handles errors and keeps the interface clean, introduce an emergency drone to demonstrate priority routing.